

Review

Review of image-based analysis and applications in construction

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ABSTRACT

Image-based analysis techniques offer a robust way to solve engineering problems due to the availability of visual data (e.g., surveillance cameras). Hence, research efforts have focused on applying Image-based techniques in the construction industry to improve the safety and productivity of construction operations as well as the resilience and sustainability of the construction assets. This paper explores the state-of-the-art in Image-based analysis techniques and their applications in construction. Over 100 journal papers were retrieved from the Scopus database for an in-depth review of major applications, benefits, and areas of future research potential. Accordingly, Three main research directions were identified that utilize image-based technologies: (1) construction safety; (2) progress monitoring; and (3) damage assessment. It is observed that most research efforts focused on object detection (e.g., hardhats, defects) for safety inspection and repair planning. Potential future developments include integrating object detection with quantification and sizing techniques to develop more comprehensive applications.

1. Introduction

The construction industry is the main contributor to developing and maintaining the civil infrastructure, with billions of dollars spent every year. However, the construction industry has been deemed to be relatively conservative in terms of adopting data-driven technology innovations to improve safety and productivity [1]. In contrast to other industries, productivity in construction has not improved over the past decades [2,3]. The U.S. construction-sector productivity is lower today than it was in 1968 [4]. Improving construction productivity worldwide, however, has the potential to increase the added value by 1.6 trillion dollars, which corresponds to a global GDP increase of 2% [5]. The construction industry is currently one of the least digitized industries in the world according to MGI's digitization index [6]. For example, it is estimated that current site managers consume almost half their time manually collecting and processing progress monitoring data before making a decision [7]. Hence there is a clear need for "automation" to achieve maximum efficiency. Among the most promising automation technologies is Image-based analysis as it offers not only a faster and more consistent way to obtain more in-depth information than obtained from manual visual analysis.

Computer vision and image-based learning techniques have recently

matured to allow for computers to automatically analyze images and videos. Computer vision techniques have been introduced since the 1960s [8] and have attracted various researchers in multiple other fields because images are easy to collect and, in some cases, readily available (e.g., security CCTV cameras). Currently, many researchers are investigating the use of advanced image-based methods in the construction industry with potential applications to improve workers' safety, monitor activities progress and facilitate project management, assess structural damages (e.g., crack detection) for structural health monitoring and rehabilitation purposes, among others.

A major area image-based analysis can address is construction site safety. The construction process itself is very challenging due to the extensive interactions among workers, equipment, site spaces, materials, and the environment, which exposes participants to multiple hazardous situations on a daily basis, making them prone to injuries or even deaths. Injuries in the construction industry constituted 7% of non-fatal injuries and 14% of workplace deaths in the United States in 2018 [9]. In Canada, incidents in the construction industry have constituted around 10% of lost-time claims and 20% of workplace fatalities over the past three years [10]. In Great Britain, 54,000 injuries are happening to construction workers each year, the second-highest number of injuries among all industries [11]. Such accidents have cost the British economy

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1.2 Billion pounds in 2017/18 [11]. It is estimated that 80% of accidents on construction sites can be attributed to the workers' behavior [12]. This includes not wearing protective equipment such as hardhats and safety harness, being struck by vehicles and construction equipment, maintaining unhealthy postures, and others. Hence, monitoring the construction site to limit such unsafe behavior is of extreme importance. However, the current monitoring methods are inefficient as they depend on manual observations and data collection.

Another area that could potentially benefit from image-based analysis technology is productivity and progress monitoring. Out of every ten construction projects, five are behind schedule and six are over-budget [13]. Among the major reasons for such poor performance is the flawed progress monitoring and reporting which leads to improper management and decision making [14]. Traditional worksite monitoring practices are conducted manually through visual observations and using paper-based drawings or, at best, tablets that require specific views of drawings to be manually generated and rendered in advance for each inspection task [15]. This process is very time-consuming, given thousands of elements and hundreds of construction activities on the site, as well as prone to errors. For example, large constructions, such as highways, require traveling between the different work areas to gather information or attend specific issues which translates to a misuse of time and transportation costs which makes the monitoring efforts infrequent and untimely enough for effective decision making. In contrast, image-based methods can be used to provide real-time information about the status of various operations throughout the construction site through documenting work progress as well as worker's and equipment's behavior and productivity. This allows project managers to identify delays in the construction plans and make better and more timely decisions concerning the project progress moving forward.

Image-based analysis is not only useful during construction, but also after construction as well as all infrastructure components need to be continuously monitored to ensure they are physically fit and viable for usage. Even after the structure is fully constructed, frequent monitoring is required to help with decision making and resource allocation regarding the structure's rehabilitation and upgrades. Traditional methods involve sending inspectors to assess the asset conditions and file reports that estimate any required maintenance work. While visual inspections are widely used for preliminary and regular inspections because of its effectiveness in detecting external defects such as cracks and spalling which could be evaluated as symptoms of potential structural degradation, subjectivity and inconsistency are inevitable in visual inspections as different inspectors may evaluate the same structure differently [16]. Furthermore, some parts of the structure are not accessible for inspectors, making an overall assessment sometimes impractical [16]. An inspection site visit typically takes 4 h to be completed and for each hour spent in the field for inspection, additional

three hours are spent in the office to generate the reports [17]. As an example of how time-consuming inspection is (using current methods) Ontario ministry of education has issued a bid in 2010 seeking a company to inspect a total of 4800 schools over a five-year period [18]. Errors and gaps produced by in manual assessments not only waste time and money, but can also lead to catastrophic events, such as the collapse of I-35 W highway bridge in Minneapolis, MN which caused 13 deaths and 145 injuries [19]. or the failure of Oroville dam spillways which took place in February 2017, forcing around 200,000 Californians to evacuate [20,21].

With image-based analysis being a potential key tool in enhancing the construction industry, this paper reviews previous works regarding the use of image-based analysis in the construction industry, following the research methodology in Fig. 1. It highlights current research trends and innovative works, challenges and limitations, and future research potential. The first step of the research technology (section 2) presents the approach used in collecting the journal papers to be analyzed. The second step (section 3) is presenting a general overview of the findings, including a thorough explanation of the most commonly used computer vision techniques. Finally, the papers were further analyzed according to their different applications to highlight notable achievements and major research gaps (sections 4, 5).

2. Survey methodology

An extensive literature review was conducted using the Scopus journal database as it automatically aggregates the search results from multiple academic databases, including: the American Society of Civil Engineers (ASCE) library, Elsevier (ScienceDirect), Wiley, IeeeXplore, and others. Only high-level keywords ("Computer vision" OR "Image*" OR "Video*") AND ("Construction") were used so that a wide search area can be covered. To highlight recent technical advances, the search was only limited to journal papers published in or after 2015. Results from the search query (9000 papers) were further filtered by manually going through the papers' titles and abstracts to determine the relevancy of the articles. Resulting articles (total 100) were quantitatively analyzed in terms of country of origin, the technique used and the application for which it was used. Finally, the papers were further analyzed based on their application areas to highlight the comments, notable contributions, and discussions brought forward by the authors.

3. General survey results

Analyzing the 100 papers found in the construction literature related to image-based analysis, Fig. 2 illustrates an overview analysis of the main research areas and their applications. Main applications of image-based analysis in construction can be grouped into three main

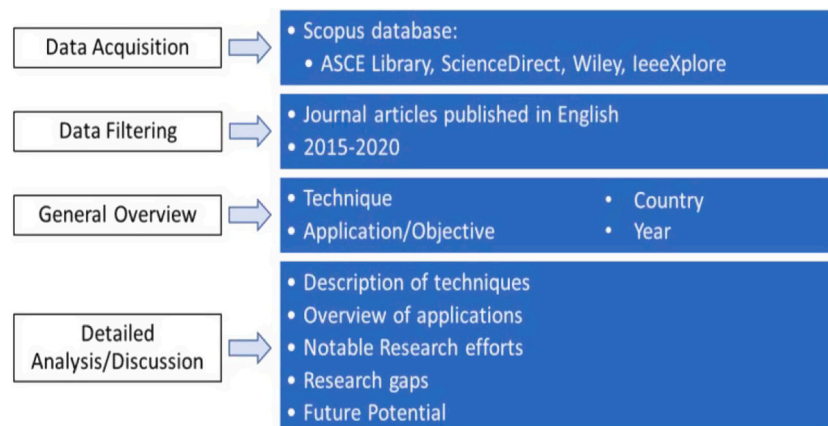


Fig. 1. Outline of the research framework.

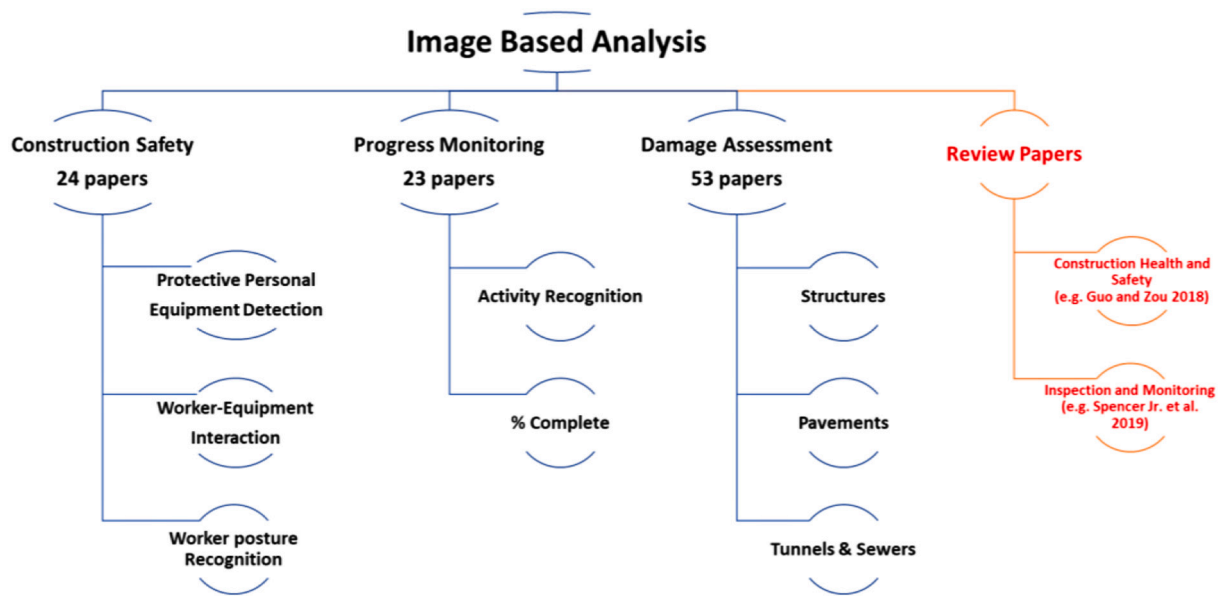


Fig. 2. Main research areas and applications of image-based analysis in construction.

categories; (1) construction safety; (2) progress monitoring; and (3) damage assessment. A number of related review papers were also investigated (right-hand-side of Fig. 2).

Before undergoing further analysis, the number of publications per year, both in total and according to each of the three main categories shown in Fig. 2, is shown in Fig. 3. It can be seen that there is an increasing trend in using Image-based analysis techniques in all three main categories over the past years, except in 2019 which has seen a drop in the papers tackling damage assessment issues which in turn affected the overall count. Papers published in 2020 have not been part of creating this chart as the comparison would be unfair since, at the time of writing this paper, only two months of the year have passed.

Grouping the papers by the country of origin based on the affiliation of the first author is shown in the heat map in Fig. 4. All 100 papers come from a total of 16 countries with almost 50% being contributed by China (24%) and the United States (23%) alone. The next major contributor is Canada (13%), followed by Korea (11%) and Hong Kong (10%).

In terms of the range of the available image-based analysis techniques, a mind map highlighting the major techniques used in the reviewed papers is shown in Fig. 5, representing four categories: Object detection/Classification (identify object and its condition); Quantification (object size); Object tracking (object movement over time); and Edge detectors (object boundary lines). Out of the four categories, object

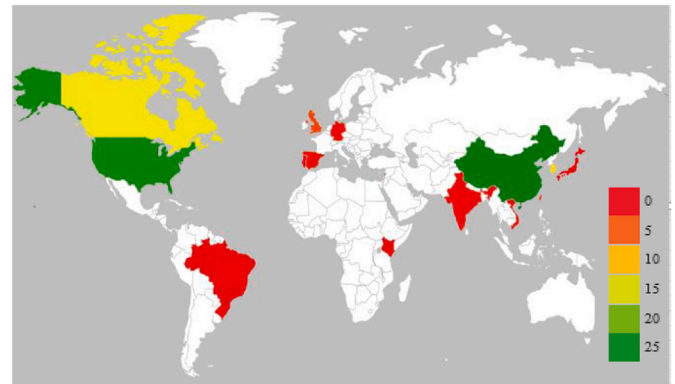


Fig. 4. Geographic heat map representing the number of publications per country.

detection and classification techniques where the most focused on by researchers, with convolutional neural networks (CNNs), an advanced variation of artificial neural networks (ANNs), being the most commonly used (50 times). This has been followed by feature extraction algorithms (20 times), and support vector machines (10 times). Object

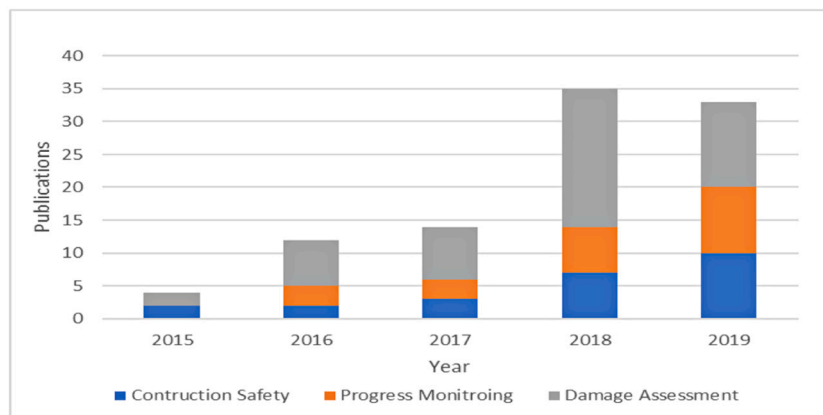


Fig. 3. Number of publications addressing a certain research category by year.

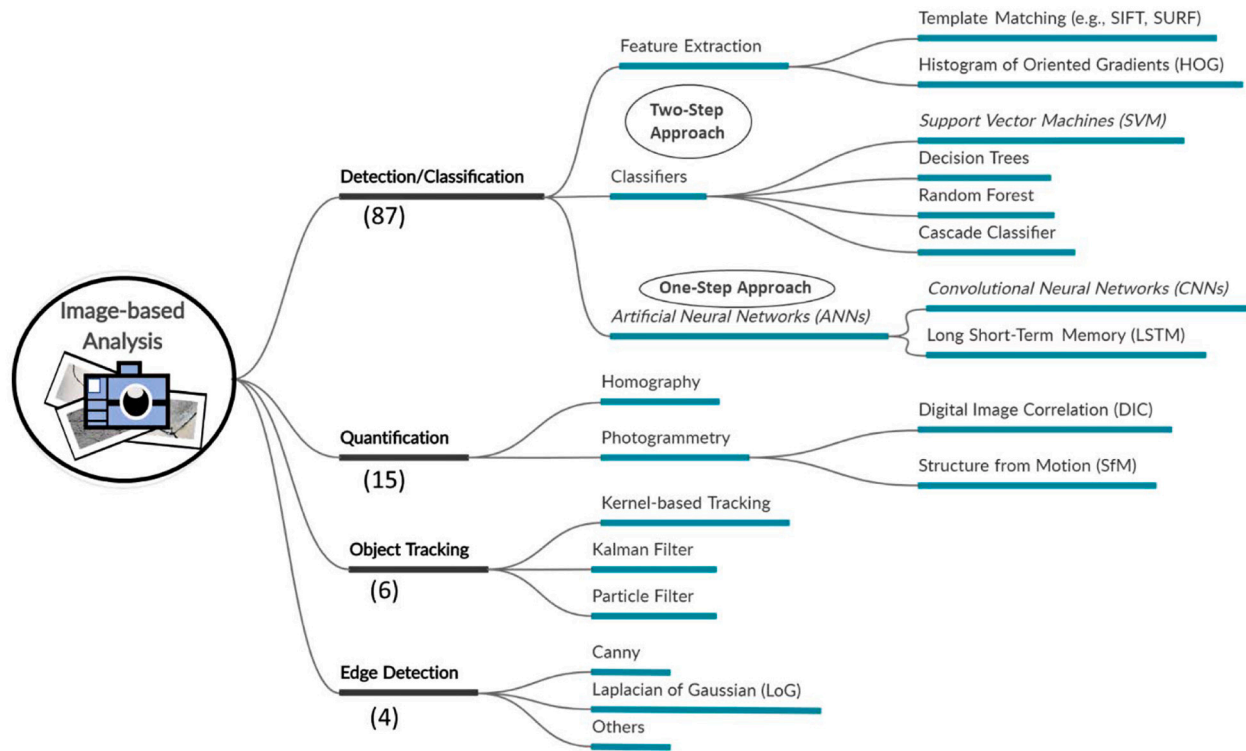


Fig. 5. Mind map of existing image-based analysis techniques.

quantification and sizing techniques were only used in 15 instances. Most of the time photogrammetry was used (13 times) as it is more robust, while homography/image-based measurement was used only twice. Object tracking and edge detection algorithms were used only six and four times, respectively.

3.1. Description of image-based analysis techniques

A brief description of the image-based analysis techniques presented in Fig. 5 is presented below.

3.1.1. Object detection and classification

These techniques can detect objects of interest from images and videos as well as classify them into predefined categories (e.g., horizontal vs. vertical crack). Intuitively, to be able to classify objects through images, first, there is a need to detect them. Such detection typically takes place by recognizing the objects' distinctive features. Therefore, object detection and classification is typically a two-step approach; detection then classification. Some of the commonly used techniques are:

Feature Extraction Algorithms: a feature is a distinctive characteristic of the object. Feature extraction algorithms use such distinctive feature(s) (or object template) to detect objects of interest inside cluttered scenes. An example is in Fig. 6: where an object (a rectangular card) is being detected inside a larger scene. Available feature extraction and template matching algorithms include: Scale Invariant Feature Transform (SIFT); Speeded Up Robust Features (SURF); and others.

Histogram of Oriented Gradients (HOG): A way to describe the images' distinctive features as a distribution of local intensity gradients or edge directions, even without precise knowledge of the corresponding gradient or edge positions [23]. Hence, the image is divided into cells of a predefined size, then the occurrences of gradient orientation in each cell are counted to build the descriptor vector. Finally, normalization is performed to regulate the variability in the image [24]. HOG pays huge attention to the shape of the detected object. As such, [25] used HOG for pothole detection and accordingly developed an automated system by

combining HOG with a Naïve-Bayes classifier achieving 90% accuracy on a 120-image dataset.

Support Vector Machines (SVM): While not used exclusively for computer vision purposes, SVM is one of the most powerful classification techniques [26]. Despite being inherently a two-class classifier, SVM can also be effectively applied for multi-class classification problems [26]. The SVM classifier can find an optimal hyperplane that separates data into two classes. In essence, SVM can be thought of as a high-dimension discriminant analysis, where the objective is finding the thresholding function that achieves the best binary classification of data.

Artificial Neural Networks (ANNs): ANN is a computing system that is primarily inspired by the human nervous system, the structure of the brain, and the way humans learn from examples. What sets ANNs apart from the other detection/classification techniques is that ANN integrate both detection and classification steps into its design and architecture. Since 2012, performance of ANNs has significantly improved because of the developed novel architecture that included multiple hidden layers. As such, the "new generation" of ANNs is now more commonly referred to as Deep Neural Networks (DNNs). In the image analysis domain, one of the recently introduced and most important types of DNNs is the convolutional neural networks (CNNs). An example of a CNN is shown in Fig. 7: where a handwritten text is classified into the correct digit. As seen in Fig. 7, CNNs consist of a combination of convolution (examining the image using a small matrix that moves over all image pixels) and pooling operators that extract the notable features of the input image and pass them on to a group of fully connected layers that classify the image based on its features. To reduce computational time especially when trying to detect multiple objects in the same image, [27] proposed a method where specific regions of the image are selected to be analyzed instead of analyzing the entire image, naming the new technique R-CNN (CNN with region proposals). Such breakthrough has led to the development of faster R-CNN algorithms such as the YOLO (You-Only-Look-Once) [28] and SSD (Single Shot Detector) [29] where only one image is needed to detect multiple objects, making them more practical for real-time applications. Other ANN variations include residual neural networks where some layers can be skipped for faster

processing, and Long Short-Term Memory (LSTM) networks which can analyze actions from data sequences (e.g., video frames) to obtain temporal information [30].

3.1.2. Object quantification

In the literature, two main methods are used to obtain real-life measurements from images: homography, and photogrammetry. These are as follows:

Homography: The homography maps two different images of the same object recorded from two different viewpoints as shown in Fig. 8. While the four points on the original object form orthogonal lines, this is not the case in either image due to the distortion caused by the camera lens. Such distortion can be rectified as there exists a homography that relates the image to the real-world object [31]. Consequently, homography can be used to directly obtain real-life measurements from captured images.

Photogrammetry: Photogrammetry, often referred to as Structure from Motion (SfM) or Digital Image Correlation (DIC), uses feature detection and homography to reconstruct a 3D model of an object from 2D images. It is widely used in mapping, surveying, and historical preservation applications [e.g., 32]. The technique relies on feature matching between images to estimate the camera positions and consequently generate a 3D point cloud. Hence, it is recommended that a minimum overlap of 60% must be present between images to achieve good results [33]. With the proper scale, real measurements can be obtained either directly from the point cloud or by using triangulation.

3.1.3. Object tracking

In addition to object detection and classification, object tracking provides important information in terms of how the object changes its posture or location, for example, over time. This is important for progress monitoring and productivity analysis purposes. The objective of object tracking is to estimate the future state of an object based on all the measurements taken up to that point [34]. The following techniques are the most commonly used for object tracking.

Kernel-based Tracking: As the name suggests, the object of interest is represented by a special window (isotropic kernel) [34]. Hence, the tracking/localization problem becomes a search problem for a “similar kernel” in the subsequent frames. Such search problem is solved using a similarity function and a gradient optimization method [34].

Kalman Filter: Kalman filters offer the ability of not only tracking an object across multiple picture frames, but also predicting its location in future frames as well. Instead of the kernel representation used in the previous method, the state of an object is expressed as a vector including its position, velocity, and acceleration among the x, y, and z axes. Hence, a state transition matrix is defined which predicts the object location in the next frame [35]. The actual object position is then compared to the prediction to adjust the transition matrix to further improve its predictions.

Particle Filter: Particle filtering is a robust technique that is able to overcome the problem of obstructions when it comes to object tracking. This method includes two main phases: first, the object of interest is initiated using a rectangular window, similar to the kernel-based tracking method, and hundreds of particles are generated inside that window; then the particles are propagated in the next video frame and resampled based on the likelihood of the particle being observed to generate a new window. This way, the object of interest can be tracked across multiple frames [36].

3.1.4. Edge detection

Edge detection refers to the use of special filters to detect edges in an image, such as a crack in a deteriorated bridge deck [37]. Edges are detected based on discontinuities in image color and/or brightness. One of the most commonly used edge detectors is the Canny edge detector (resulting in the binary image in Fig. 9b). Part of the process includes setting two thresholds, a minimum and a maximum, are then set. If the pixel value is above the maximum threshold, it is considered a “strong edge”. If it is below the minimum threshold, it is not an edge. If it is between the two thresholds, it is considered an edge only if it was connected to a strong edge. [37] compared the performance of six different edge detectors against a Deep Convolutional Neural Network (DCNN) by applying them to a 100-image dataset of concrete panels for crack detection purposes. The experiments have shown that DCNNs can detect finer cracks with higher accuracy, with the LoG filter being one of the most efficient filters in terms of both accuracy and computation time.

4. Applications of image-based analysis in construction

A perfectly automated inspection or monitoring framework consists of two main modules; automated data acquisition using UAVs or stationary cameras (e.g. CCTV), and automated decision making through data processing using image-based analysis techniques [8]. While the rapid growth in the drone industry has made the first module a reality, the analysis module and turning images into decision-support information remains a challenge. Towards this goal, the following three subsections review the literature in the three major application areas featured in Fig. 2, summarizing the existing body of knowledge based on the reviewed papers and highlighting notable contributions.

4.1. Construction safety applications

A sizeable portion of the reviewed papers (24%) has focused on using image-based analysis technology to address construction safety. The reviewed literature extensively reported the great potential benefits in terms of improving workers' safety on construction sites. Table 1 summarizes the reviewed literature. In addition to the reported accuracy of the developed model (column 17), the table is divided into three main

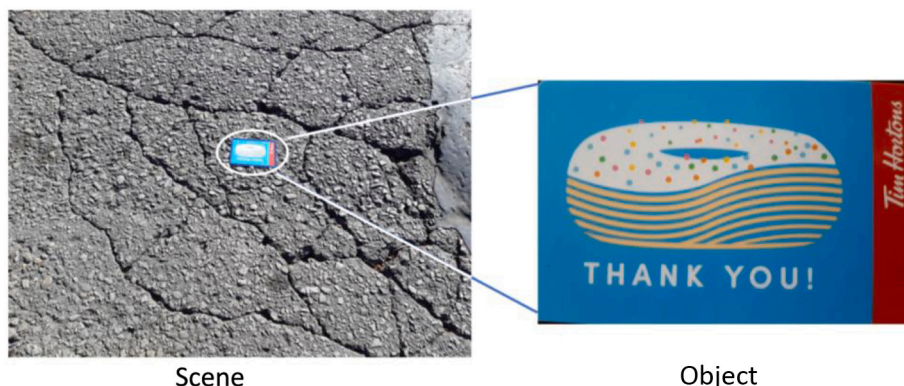


Fig. 6. Example scene with an object with distinctive features (rectangular card).

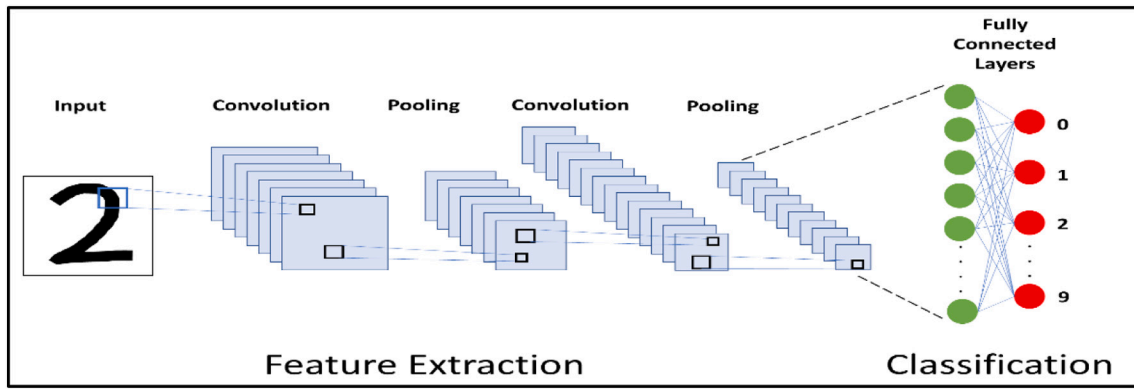


Fig. 7. Example of convolutional neural network architecture.

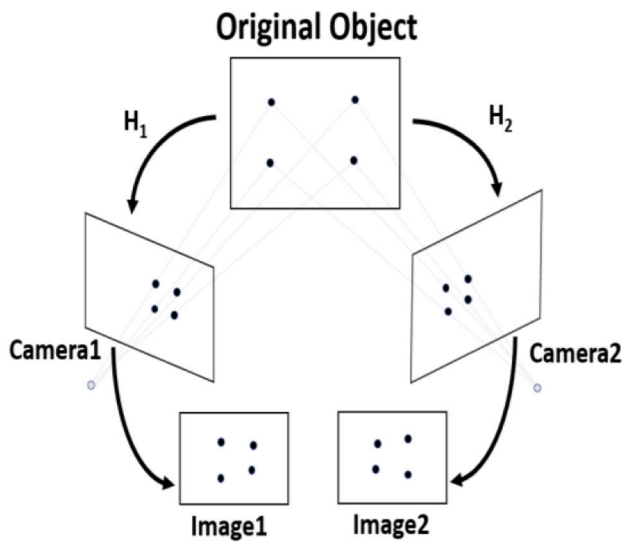


Fig. 8. Multiple images of an object can be used for direct measurements.

sections:

- the technique used by the researcher (object detection, quantification, object tracking, and edge detection), with the three most popular object detection techniques (ANN, feature extraction and SVM) featured in columns 2–4
- the object being detected (workers, equipment, or others); and
- the safety issue being addressed by the research (e.g., posture and ergonomics, hardhats and PPE negligence, worker/equipment

accidents). Three of the most instead of grouping them under the object detection category.

Most of the research efforts were geared towards workers detection and tracking to identify major safety concerns such as the absence of hardhats and safety harnesses (example in Fig. 10) as well as awkward postures. This was expected as most construction safety issues stem from workers' behavior [12]. Other researchers tracked the movement of equipment to prevent worker-equipment collisions [46], while others aimed to detect a different safety hazard such as the presence, or rather absence, of guardrails [51].

According to Table 1, the developed models have a satisfying performance level (91% on average). An interesting observation in Table 1



Fig. 10. Example of detecting workers, hardhats, and safety harnesses.

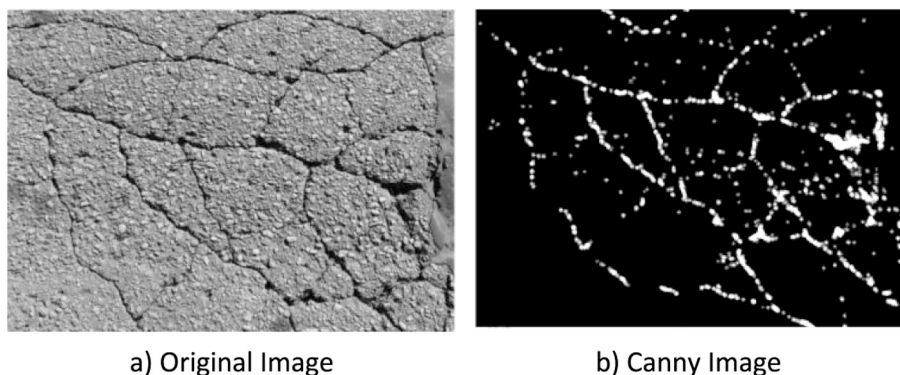


Fig. 9. Canny edge detection.



Fig. 11. Equipment activity classification.

is the sudden rise in the use of ANNs technique over the past few years. ANNs have not been used to investigate construction safety issues from images before 2018 and then, due to the CNN developments such as the release of google Tensorflow 1.0 in 2017 [38], became the method of choice for almost all researchers instead of the relatively simplistic feature extraction algorithms. For example, the same task (detecting hardhats) was tackled by [39,40] using feature extraction algorithms such as HOG and cascade classifiers, but [41,42] used ANNs and managed to achieve higher accuracy levels. The advances in image-based analysis techniques have allowed researchers to investigate more complex issues such as assessing workers postures to protect them from developing work-related musculoskeletal disorders (MSDs), an issue that requires complex computations in order to develop and analyze workers' 3D skeleton models from ordinary 2D images. As such, five of the six papers attempted to tackle the issue of ergonomics (the five papers published after 2017) used CNN architecture.

Despite the seriousness of falls from heights in construction sites, causing a third of construction worksite fatalities [61], not a significant amount of research has attempted to address this issue compared to other safety concerns such as the presence of hardhats, accidents, or worker postures. Column 16 highlights some interesting research that did not directly address a safety issue. For example, [55] focused on detecting unauthorized work and [43] used CNNs and feature extraction algorithms to identify workers on site. Some research contributions did not directly tackle a safety issue, but rather provide the building block for future applications. For example, [53] developed a method to match construction workers across multiple scenes based on appearance and geometry.

While the image-based research related to construction safety provided many interesting potential applications, there is a need to analyze the impact of the detected actions of workers and equipment from images (e.g., speed, posture, proximity, etc.) on higher-level behaviors and decisions such as preventing potential collisions or improving design quality [22]. Such analysis would provide more than mere measurements, but smarter decision support.

4.2. Progress monitoring applications

This research direction has been pursued by a sizeable portion of the reviewed papers (23%), extensively reporting the great potential benefits of using image-based analysis to monitor the progress of activities on the construction site. Table 2 summarizes the reviewed literature, which is divided into two main sections. The first section of the table focuses on the techniques used (similar to Table 1), while the second section highlights the different objects recognized during progress monitoring (workers, equipment, or others).

Similar to the techniques used in analyzing site safety, the most used technique (especially in works developed in or after 2018) is ANNs. Most of the Research in the area of progress monitoring can be categorized based on what is the object being tracked; worker or equipment (Columns 9 and 10, respectively), with both getting equal attention from researchers. The goal of the tracking is to determine the time spent by the

equipment or the worker doing a certain activity (example in Fig. 11). An example of tracking the performance of construction equipment is the work of [68] who were able to develop a model capable of detecting different equipment activities from video sequences by integrating CNNs, hidden Markov models, Gaussian mixture models, and SVM classifiers. An example of tracking the workers' performance is the work of [63] who used Long short-term memory (LSTM) neural networks to identify workers in a construction scene and categorize them into working groups based on the activities being performed.

Other researchers tracked neither workers nor equipment (column 11). [7], who used SVM and edge detection to calculate the percentage complete of tiling works based on the detected area of constructed tiles as compared to the corresponding BIM model [78]. developed an algorithm that would detect the state of construction of interior drywall activities (e.g. studs installed, drywall plastered, etc.). [70] used worker detection and tracking technology to detect the different onsite work-spaces (e.g., working areas, rest areas, laydown areas, etc.).

The average accuracy for the reviewed models in this category is 85%, the lowest of all three categories. One reason for that drop in performance level is the problem of occlusions which makes the models estimate the workers and equipment pose and location incorrectly. This explains why the models developed to track stationary objects such as material for percentage complete [e.g., 7,78] have higher accuracies (88% on average) than the other models that tracks workers and other moving equipment. Among the proposed solutions to resolve the issue of occlusions is the use of multiple cameras and/or place them at locations with a bigger field of view [77], or rely on techniques that track objects across multiple frames of a video feed and use stochastic modeling to predict the object position when occluded [35,36].

Continued work in this research direction can enhance project progress and resource monitoring practices. As mentioned above, one major technical challenge to this research direction is the fact that image-based analysis technology can only analyze what they can "see", which casts a shadow of doubt on their ability to fully monitor the construction site 24/7 and under different weather conditions. Also, maintaining good quality images over the entire site would require installing more cameras which is costly especially in linear projects such as highways. Furthermore, installing more cameras may affect the morale of the workers by making them feel "watched" which would, in turn, affect their productivity [82].

4.3. Damage assessment applications

Based on the conducted review, using image-based analysis to analyze the current condition of assets has been a major research direction, addressed by most of the reviewed papers (53%). The reviewed literature extensively reported the great potential benefits in terms of assessing damages in roads, buildings, and other components of the built environment. Table 3 of the analysis summary is divided into three main sections:

Table 1
Summary of image-based analysis applications in construction safety.

Technique								Recognized Object			Safety Issue					
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)
Reference	ANN (e.g., CNN, LSTM)	Feature Extraction	SVM	Quantification	Object Tracking	Edge Detection	Others	Worker	Equipment	Other (e.g., guardrails)	Hardhats	Accidents	Falls	Ergonomics	Other (e.g., security)	Accuracy (%)
[43]	x							X							x	79
[44]	x							X								N/A
[45]	x							X							x	N/A
[46]		x						X			x					90
[47]	x							X						x		94
[48]	x							X	x			x				96
[49]	x							X						x		84
[42]	x							X			x					91
[50]	x			x				X	x			x				90
[51]	x							X					x			90
[52]				x				X							x	95
[41]	x							X			x					95
[53]	x									x					x	96
[30]	x							X						x		95
[54]	x							X						x		94
[55]	x							X					x			83
[56]	x							X							x	97
[57]			x					X						x		98
[40]		x						X			x					77
[58]		x							x			x				88
[59]					x			X	x			x				95
[35]					x			X	x						x	N/A
[39]		x						x			x					93
[60]		x						x				x				93

Table 2

Summary of image-based analysis applications in progress monitoring.

(1)	Technique							Recognized Object			
	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Reference	ANN (e.g., CNN, LSTM)	Feature Extraction	SVM	Quantification	Object Tracking	Edge Detection	Others (e.g., K-means)	Worker	Equipment	Other (% complete)	Accuracy (%)
[7]		x								x	N/A
[62]	X								x		82
[63]				x						x	N/A
[64]				x						x	91
[65]	X							x			95
[66]	X								x		60
[67]	X								x		94
[68]	X		x				x		x		87
[69]				x				x			96
[70]	X						x			x	81
[71]	X								x		96
[2]	X							x			85
[72]				x						x	93
[73]	x							x	x		93
[74]	x							x			81
[75]		x								x	93
[76]	x							x			80
[77]						x			x		70
[78]			x							x	85
[79]		x	x					x	x		95
[36]					x			x	x		78
[80]			x				x	x			59
[81]		x							x		87

- the technique used by the researcher, this followed the same procedure used in [Tables 1 and 2](#) except for adding the “object Tracking” techniques to the “others” category because of the low number of research papers utilizing this category of techniques;
- The type of the structure being inspected (e.g., vertical structures, roads, sewers, etc.);
- The type of defect (e.g., cracks, displacements, etc.); and
- The reported accuracy of the developed models.

Overall, the developed models have reported a satisfying level of accuracy (90%). The same observation highlighted in [Tables 1 and 2](#) (the popularity of ANNs) is also observed in [Table 3](#), with early adopters of ANNs in 2017 [e.g., 126, 127]. Looking into [sections 2 and 3](#) in [Table 3](#) shows that most of the ongoing research is geared towards crack detection in vertical structures such as buildings and bridges. For example, [111] developed a crack detection algorithm that is six times faster than its conventional counterparts without sacrificing accuracy. Their interesting approach processes the image to extract vertical and horizontal cracks separately then fuses them to produce a final image where the cracks are highlighted. [127] used CNNs to detect concrete cracks that achieved 98% accuracy for both training and testing sets, before expanding the detection to steel corrosion, bolt corrosion, and steel delamination [119]. To achieve this impressive accuracy, however, a 40,000 training-image-dataset was used. To utilize a small data set, [89] reached high accuracy by first extracting the features using conventional CNN, then using feature fusion to obtain higher precision. [85] combined CNN with a geotagging system for instant damage identification and localization.

Another prominent area of research is pavement defect detection (column 9). [118] proposed a pothole detection and localization scheme relying on the SIFT feature extraction algorithm and support vector machine. Testing the scheme has shown its capability of pothole identification with 95% accuracy. [106] proposed a novel CNN architecture, CrackNet, for automated pixel-level crack detection on asphalt surfaces. An experiment using a 200-image testing dataset showed that CrackNet can achieve high Precision (90.13%) and Recall (87.63%) simultaneously. Furthermore, CrackNet can be used in conjunction with the data collection software because of its capability to utilize parallel

computing techniques. Examples of works in other inspection “sub-domains” shown in columns 10 and 11 include the ANN-based model developed by [16] for detection of moisture marks in subway networks, the CNN-based schemes developed by [101,108] to automatically detect cracks, deposits, and root intrusions in sewers from CCTV inspection videos, and the use of support vector machines by [99,134] to detect loosened bolts in steel structures.

While the ongoing works provided notable efforts towards defect detection, there are still grounds to cover in terms of fully automating the structural inspection process. For example, efforts towards defect measurements and quantification (column 5) have not been pursued extensively the research community although they are important for decision making (e.g., [86,105]). Furthermore, most existing research adopts a binary classification/detection approach (defect versus no defect) without investigating the level of severity of the defect detected. In addition, many studies depended on lab experiments and photos of smooth and clear surfaces, which are more ideal than outdoor conditions.

4.4. Other applications

This section discusses other notable applications for image-based analysis that either did not fit in one of the three specific categories discussed earlier.

4.4.1. Image-based model reconstruction

Using techniques such as homography and photogrammetry discussed earlier, researchers were able to rely on images to create parametric models of real-life objects and scenes. One of the major contributors to this breakthrough is the developments in multiple point statistics algorithms [137] which made image-based reconstruction less complex and more consistent. One of the most notable image-based model reconstruction applications is historical preservation, where [32,138] used images from drones to build a 3D model of Singosari temple in Indonesia and Christ the Redeemer statue in Rio de Janeiro, respectively. Another important application is geological modeling, [139] used high resolution panoramic imaging and higher-order statistics methods to model the Eagle Ford outcrops in East Texas.

Table 3
Summary of image-based analysis applications in damage assessment.

Technique							Structure Type				Defect Type				
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
Author	ANN (e.g., CNN, LSTM)	Feature Extraction	SVM	Quantification	Edge Detection	Others (e.g., Decision Trees)	Vertical Structures	Pavement	Sewer	Tunnel	Crack	Displacement	Multiple	Others (e.g., strains, load)	Accuracy (%)
[83]	x								x				x		65
[84]		x								x		x			80
[85]	x						x				x				78
[86]				x			x					x			N/A
[87]				x			x							x	N/A
[88]	x						x				x				92
[89]	x						x				x				90
[90]		x					x					x			N/A
[91]	x								x				x		96
[92]	x						x				x				90
[93]	x						x				x				76
[94]				X			x				x				N/A
[95]	x						x						x		92
[96]	x							x			x				78
[97]	x						x				x				91
[98]	x						x						x		99
[99]			x			x	x							x	91
[100]		x				x	x					x			97
[101]	x								x				x		86
[102]	x	x	x			x	x							x	85
[103]			x			x		x			x				96
[104]						x	x						x		N/A
[105]				x			x					x			N/A
[106]	x							x			x				90
[107]	x							x			x				90
[108]	x								x				x		83
[109]	x									x			x		87
[110]		x							x				x		80
[111]		x					x					x			N/A
[112]	x						x						x		98
[113]				x			x					x			N/A
[114]	x									x				x	95
[115]	x							x			x				92
[116]	x	x	x			x	x							x	83
[117]					x	x	x				x				94
[118]		x	x					x						x	95
[119]	x						x						x		88
[16]	x									x				x	96
[120]						x	x					x			N/A
[121]		x									x				95
[122]				x				x			x				N/A
[123]	x							x			x				87
[124]						x		x						x	92
[125]				x			x							x	N/A
[126]	x						x				x				N/A
[127]	x						x				x				98
[128]				x						x				x	N/A
[129]				x			x						x		94
[130]					x		x						x		N/A
[131]			x				x							x	86
[25]		x							x						90
[132]						x	x				x				92
[133]		x				x	x					x			N/A
[134]		x	x				x							x	92
[135]			x						x				x		97
[136]		x			x		x				x				96

In Construction, image-based model reconstruction has been used in multiple domains. For example, [140] generated a virtual construction scene using images from on-site cameras while avoiding repetitive counting of resources (e.g., the same equipment appears in images from multiple cameras) by a matching method that identifies the different onsite resources. [141] used drone images to build a model of an earthwork construction site and were successful in using the model to track earth volume change to document the progress of the different earthwork activities. [142,143] were able to correlate the captured on-site images to their respective objects in the BIM model, thus creating an up to date as-built model based on BIM's parametric information. To acquire more information from collected images, [144] proposed using thermal imaging as opposed to conventional cameras as a way to resolve issues arising from low-quality images and to better identify the different construction elements based the different material temperature.

Researchers have used image-based model reconstruction not only to model construction sites, but also to model the structures themselves. This work has been done successfully and at different stages of the building life cycle. During construction, [7,78] used images to reconstruct an as-built BIM model to track the progress of multiple different activities. Post construction, researchers have used images to reconstruct an "as-is" BIM models for existing buildings [e.g., 145]. Other researchers aimed to use images and BIM models to construct an "as-damaged" model to identify the level of damage of buildings after being impacted by natural disasters. [146,147] developed image-based reconstruction model for damage assessment of buildings affected by hurricanes and earthquakes, respectively. The model in [146] was tested using information collected during hurricane Sandy and the results revealed that, compared to a LIDAR system, the proposed model provides adequate results that help determine rehabilitation needs for the damaged buildings. While [147] used drone images to validate their model using two different case studies; a building in Japan affected by a tsunami, and a Nepalese residential district affected by 7.8-degree earthquake.

Compared to laser scanners, image-based reconstruction offers a more convenient and resource efficient alternative. However, there are still some gaps that need to be addressed in order for the technology to become more appealing to practitioners. First of all, a process to streamline the model reconstruction process is yet to be developed, which makes the technology more susceptible to user error [148]. Other sources of error in the model construction process include the quality of the collected images, the accuracy of the object detection algorithm, and (in the case of BIM models) the ability to correlate the extracted objects to the correct Industry Foundation Classes (IFCs) to properly build and update the model [145]. As such, the practicality of the technology in terms of general applicability under various circumstances and without the need for a specialist supervision is still a concern [145].

4.4.2. Physical properties evaluation

The strength and durability of a structure highly depends on the properties of the materials used. Traditionally, investigating such properties takes place through a variety of tests, some of which are destructive. However, researchers are exploring ways to study material in a non-intrusive manner through examining images of their microstructure. For example, [149] used CNNs to evaluate rock permeability and other physical properties. Similar work was performed on other material such as sandstone [150] and shale [151]. To address the problem of not being able to fully acquire all he needed images, [152] developed a stochastic process based on Markov models that would only acquire a single image to accurately model porous media such as sandstone.

This line of research greatly benefits the construction industry as it allows for a more advanced analysis of soil properties which provide the foundation for the developed structures. Another example of using this technique for the benefit of the construction industry is the ability to

estimate cement compressive strength using images of the cement microstructure, as opposed to doing compressive tests [153].

5. Summary and concluding remarks

This paper presents an overview of major image-based learning techniques utilized and their applications in the construction industry. Most applications are in the fields of site safety, progress monitoring, and damage assessment. The reviewed articles on image-based analysis applications in construction represent groundbreaking achievements in terms of using computer vision technology to enhance construction productivity and safety. To maximize the benefit from computer vision technology, there is a need for investment in high-quality cameras as images of higher resolution enable the detection of finer objects (e.g., cracks). From the survey, Convolutional Neural Networks (CNNs) are the most widely used technique by the researchers. Being a heavily training-based technique, CNNs have two main shortcomings: (1) they require supervised learning from input-output patterns, without understanding the underlying causality that correlates those patterns; and (2) they require huge training datasets to reach acceptable levels of accuracy. The absence of publicly available data for training and testing means that comparing the performance of different image-analysis algorithms is challenging (as such, the reader is advised to take the reported accuracy levels in Tables 1-3 with a grain of salt). Moreover, these research efforts seem to cater to the academic audience rather than industry professionals. The results of many of the reviewed research were obtained from controlled environments, and as such, the reported accuracy may not be reached in outdoor conditions. Future efforts should focus on the development of public databases and image datasets for the construction industry for training and testing image-based analysis algorithms. As a suggestion for future work, construction safety research could focus on developing applications that can "understand the scene" as well as incorporate temporal information for more meaningful hazard recognition. Also, progress monitoring could develop applications that are more suited to the modular and off-site construction environment. Finally, damage assessment research could develop applications that focus on detailed classification of defect severity level and quantification of defect size.

Data availability statement

All data generated or analyzed during the study are included in the published paper

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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